

Problem

A voice signal occupying the frequency band of 0.4 to 3.5 kHz is used to amplitude-modulate a 10-MHz carrier. Determine the range of frequencies for the lower and upper sidebands.

Step-by-step solution

Step 1 of 1

$$\text{Carrier frequency } (f_c) = 10 \text{ MHz} = 10,000 \text{ kHz} \quad [1 \text{ MHz} = 10^3 \text{ kHz}]$$

$$\text{Modulating frequency range, } f_{m1} = 0.4 \text{ kHz} \\ f_{m2} = 3.5 \text{ kHz}$$

So, Lower side band frequencies range, is given by

$$f_c - f_{m1} = 10,000 \text{ kHz} - 0.4 \text{ kHz} = 9999.6 \text{ kHz}$$

$$f_c - f_{m2} = 10,000 \text{ kHz} - 3.5 \text{ kHz} = 9996.5 \text{ kHz}$$

So, required range (9996.5 kHz to 9999.6 kHz)

And, Upper side band frequencies range, is given by.

$$f_c + f_{m1} = (10,000 \text{ kHz}) + (0.4 \text{ kHz}) = 10,000.4 \text{ kHz}$$

$$f_c + f_{m2} = (10,000 \text{ kHz}) + (3.5 \text{ kHz}) = 10003.5 \text{ kHz}$$

So, required range (10000.4 kHz to 10003.5 kHz)